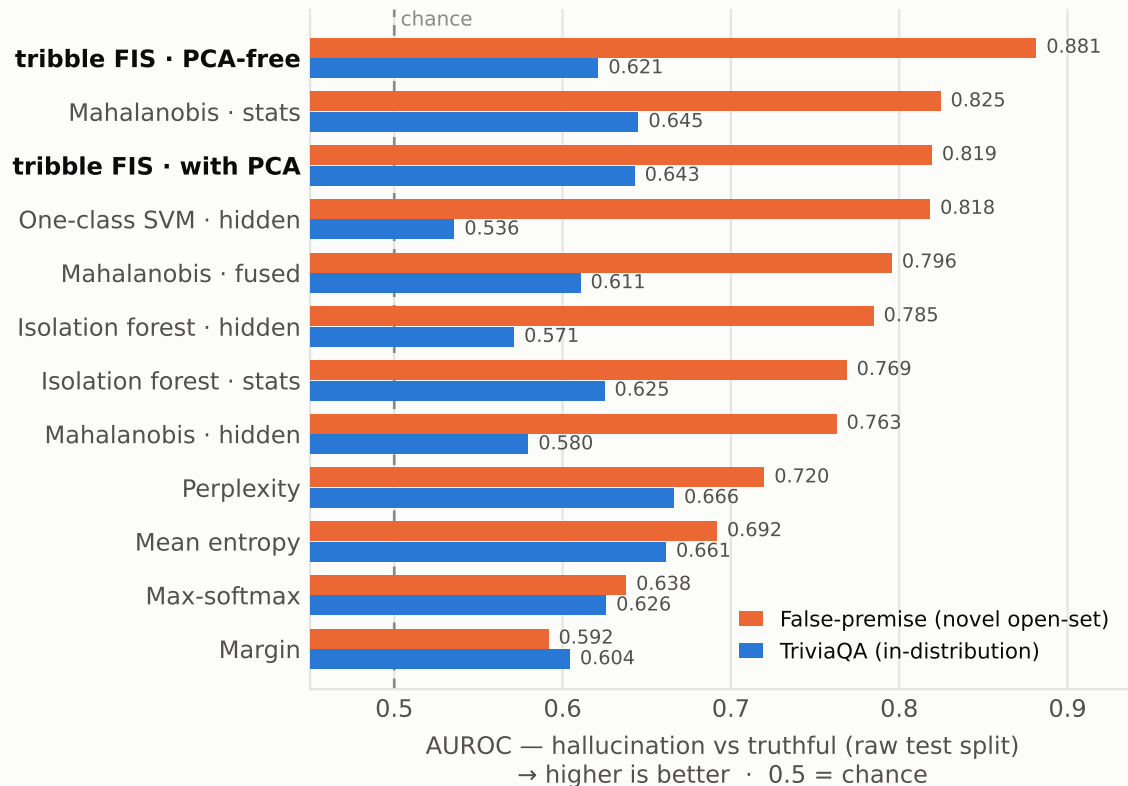


Fuzzy anomaly detection on a frozen SmolLM2-360M-Instruct

Fit on accurate (question, answer) pairs only — no hallucination seen during fitting. Answer length alone scores 0.843 on the raw false-premise split, so it must be controlled. Once it is, the fuzzy rule leads by +0.117 over the same 8 features (10/10 seeds, $p = 0.002$) while the confidence baselines fall to near chance — and the same rule on PCA features collapses below it, showing PCA was reading length.

A • Detector comparison — raw split



B • Controlling for answer length (10 seeds, mean $\pm$ std)

